

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0765

Reynolds Hill 138 kV
April 16, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0765. The Interconnection Customer proposes a 202.6 MW solar photovoltaic facility interconnecting at the Reynolds Hill 138 kV substation in the Kentucky area. The request was received on March 25, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 225 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,486,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Wixom to Albion Road 138 KV line
Contingency: North Apleton to Davisville 138 KV line
Loading: 161.7% (Rating: 371 MVA)
DFAX: 7.1%
Pre-project loading: 157.8%
2. Facility: Weber to Mullberry 138 KV
Contingency: Abajo to Fort Rucker 138 KV
Loading: 111.0% (Rating: 210 MVA)
DFAX: 49.0%
Pre-project loading: 63.8%
3. Facility: Mullberry to Salem Walk 138 KV Line
Contingency: Ft Rucker to Abajo 138 KV
Loading: 109.5% (Rating: 220 MVA)
DFAX: 49.0%
Pre-project loading: 64.4%
4. Facility: Laguna Tap to Lark Ellen 138 KV
Contingency: Abajo to Fort Rucker 138 KV line
Loading: 103.5% (Rating: 215 MVA)
DFAX: 44.2%
Pre-project loading: 61.9%
5. Facility: Weber to Lark Ellen 138 KV
Contingency: Fort Rucker to Abajo 138 KV Line
Loading: 103.5% (Rating: 220 MVA)
DFAX: 44.2%
Pre-project loading: 62.8%

6. Facility: Studio to Laguna Tap 138 KV
Contingency: Ft Rucker Substation to Abajo Substation 138 KV
Loading: 101.7% (Rating: 213 MVA)
DFAX: 44.2%
Pre-project loading: 59.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wixom to Albion Road 138 KV	Rebuild line with higher-capacity conductor	\$3,642,000	4% (\$149,000)
Weber to Mullberry 138 KV	Rebuild line with higher-capacity conductor	\$4,560,000	100% (\$4,560,000)
Mullberry to Salem Walk 138 KV	Reconductor line	\$2,724,000	100% (\$2,724,000)
Laguna Tap to Lark Ellen 138 KV	Reconductor line	\$1,406,000	100% (\$1,406,000)
Weber to Lark Ellen 138 KV	Reconductor line	\$5,396,000	100% (\$5,396,000)
Studio to Laguna Tap 138 KV	Reconductor line	\$3,251,000	100% (\$3,251,000)
	Total allocated to project		\$17,486,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.